APPLIED MACHINE LEARNING CASE

# Korea Income & Welfare

This project started as a notebook about education and income. It now works as a stronger portfolio case because the analysis is reproducible in English, the benchmark is broader, and the final conclusion is more honest about what education can and cannot explain by itself.

**89,935**

Valid records used in the upgraded analysis layer

**0.628**

Best $R^2$ from the multivariable random-forest benchmark

**8.2k USD**

Average income among doctorate holders in the cleaned sample

## Purpose and hypothesis

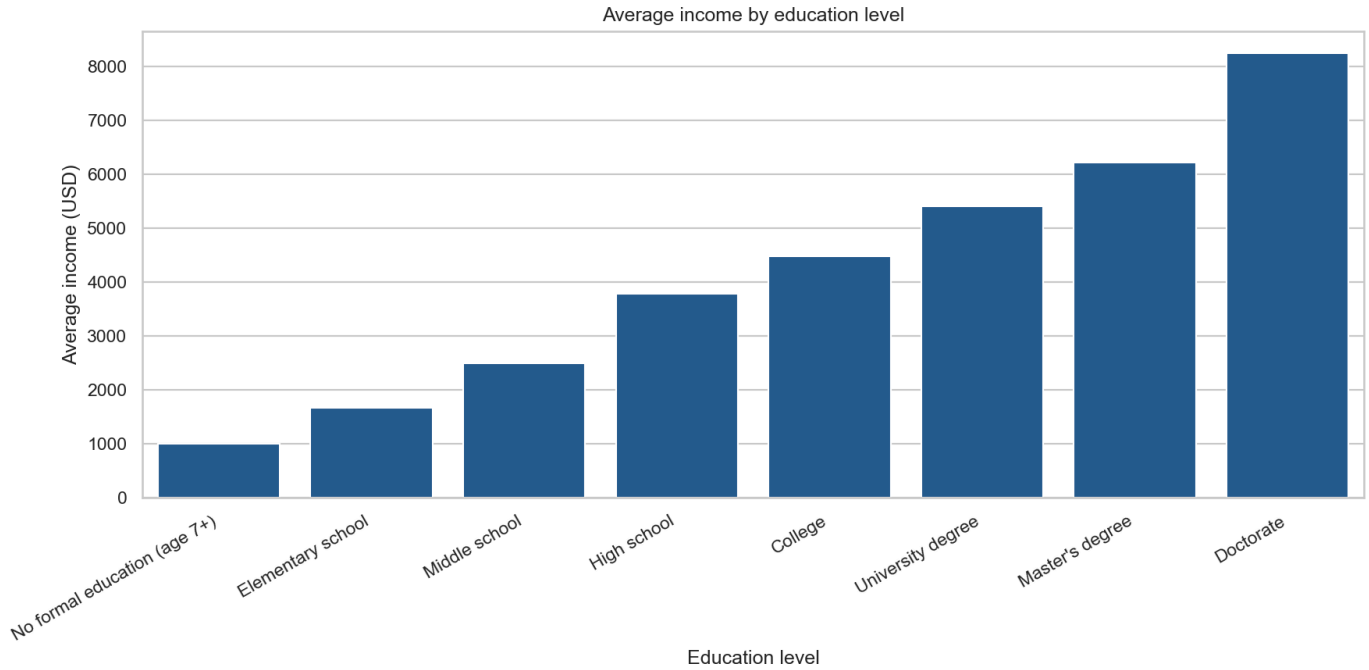
The repository asks whether education alone is enough to explain income differences in Korean welfare microdata, or whether a more complete socioeconomic model changes the interpretation.

> Main hypothesis: education is clearly associated with income, but household structure, age, and demographic context should materially improve predictive performance.

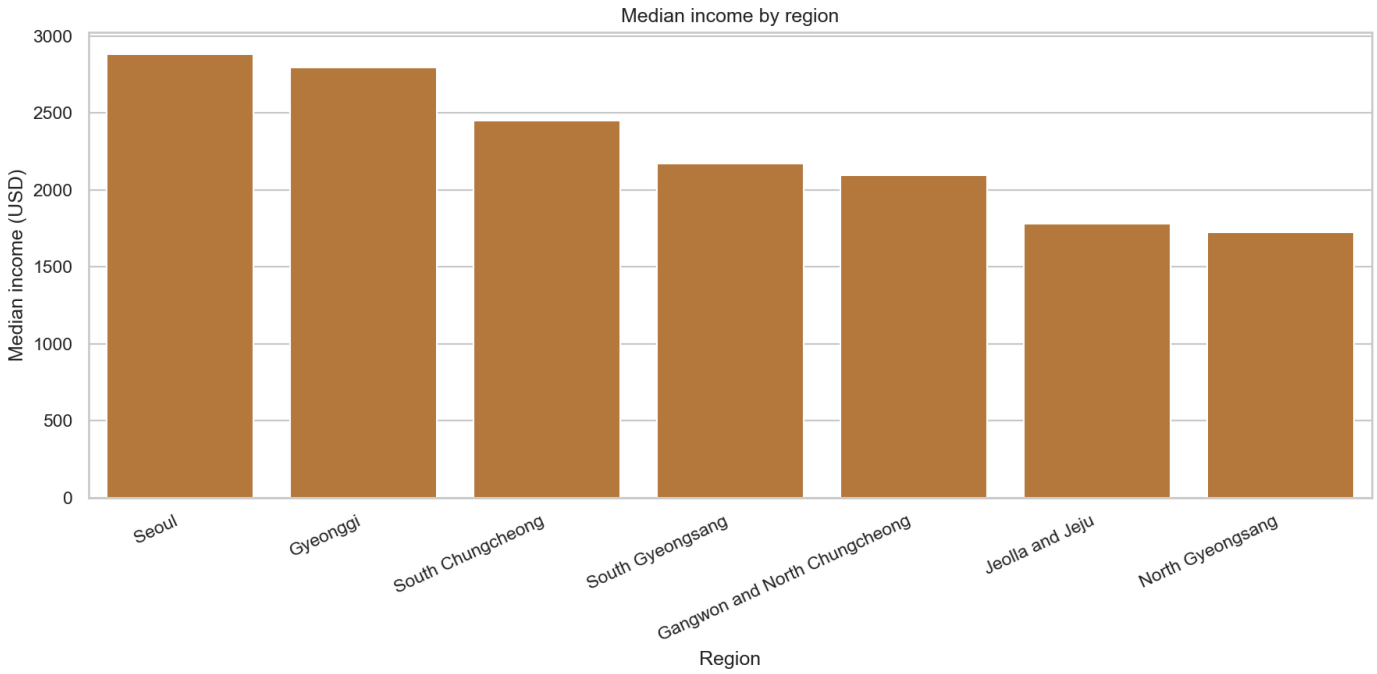
- An education-only model should capture real signal, but not enough explanatory power for a mature portfolio claim.
- A multivariable model should outperform the single-feature benchmark by a visible margin.
- The upgraded result should improve the project precisely by making the conclusion less simplistic.

## What the evidence shows

The descriptive layer supports the original intuition: income rises steadily across the education ladder and Seoul leads the regional distribution. That gives the project a clear socioeconomic narrative before any model is trained.




Income increases across education tiers, which confirms the project's starting intuition and makes the education-income link easy to communicate to non-technical readers.

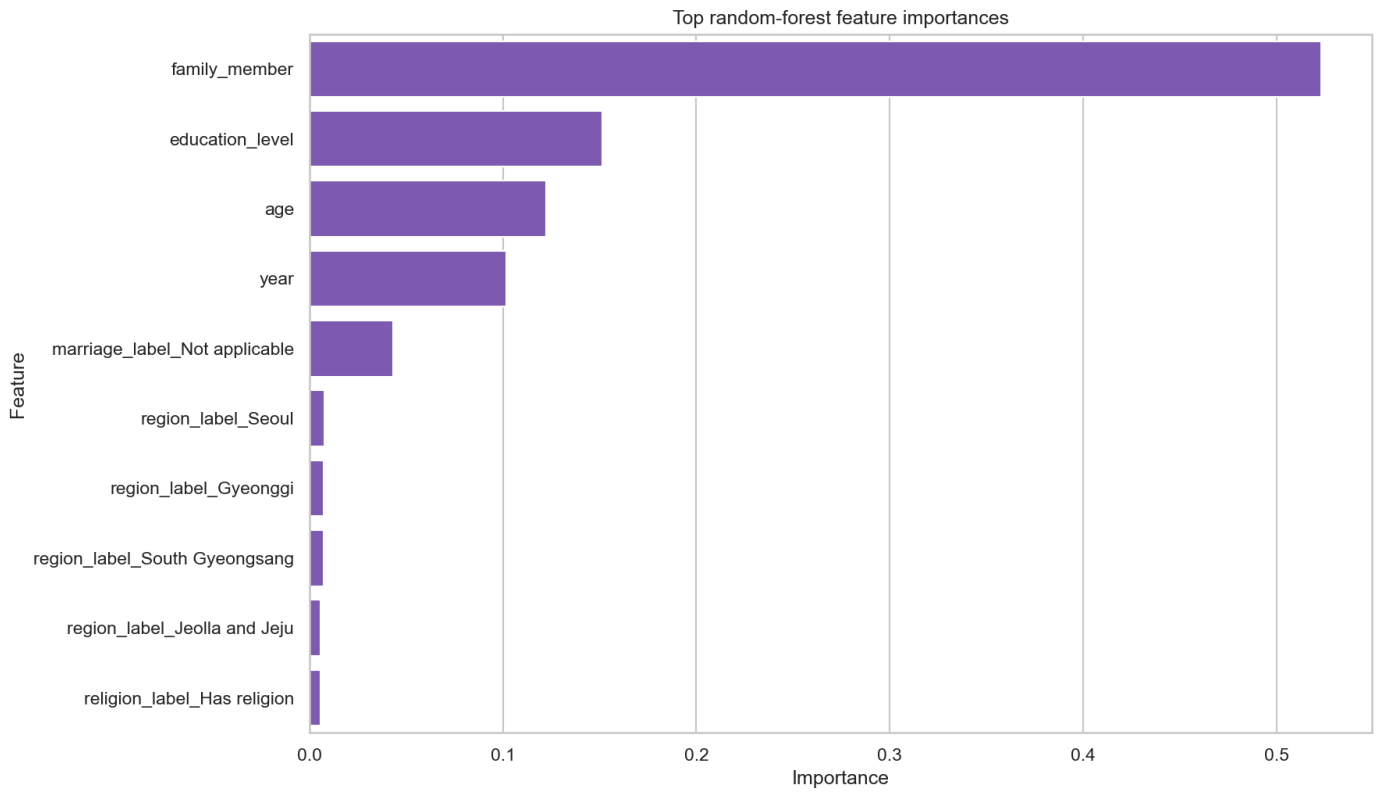



Regional differences remain visible, with Seoul and Gyeonggi leading the distribution. That matters because education is not the only context shaping earnings.

## Benchmark upgrade

The improved workflow compares three benchmarks instead of stopping at the original one-feature model.

| Model | MAE (USD) | RMSE (USD) | $R^2$ |
|---|---|---|---|
| Education-only linear regression | 1,345.40 | 1,762.56 | 0.294 |
| Multivariable linear regression | 1,049.83 | 1,414.49 | 0.545 |
| Multivariable random forest | 908.88 | 1,279.94 | 0.628 |




Family size becomes the strongest feature in the best-performing model, followed by education, age, and survey year.

## Improved conclusion

The stronger conclusion is not just that education matters. The stronger conclusion is that education matters within a wider socioeconomic structure. The original notebook found real signal, but the upgraded model shows that household structure and lifecycle effects add major explanatory power.

> Professional takeaway: this repository now reads like a more credible applied-economics case because it reports both signal and model limits instead of overselling a simple story.

That makes the project more persuasive for analytics, policy, and consulting positioning, especially when the audience wants evidence of methodological maturity rather than a single interesting chart.